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ENTERPRISE SYSTEMS DESIGN AND MODELING (WBL) - SECP3744 (01)

INTEGRATE NON-SAP APPLICATIONS WITH SAP INTEGRATION SUITE | SAP
TECHED FOR SAP COMMUNITY

**TUTORIAL : CONNECTIVITY TO NON-SAP APPLICATIONS USING SAP
INTEGRATION SUITE**

PREPARED BY : INDIVIDUAL

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Introduction

SAP Business Technology Platform (BTP) is an integrated environment that enables developers to extend and integrate SAP and non-SAP applications. For this project, I focused on the SAP Integration Suite, specifically exploring how to bridge a third-party HR system (BambooHR) with SAP BTP using the Open Connectors capability.

Tutorial Selection

Video Title: Integrate Non-SAP Applications with SAP Integration Suite | SAP TechEd for SAP Community

Source: YouTube (official SAP Community Channel)

SAP BTP Tutorial: Connectivity to Non-SAP Applications Using SAP Integration Suite.

Objective: To fetch employee data from an external system (BambooHR) and expose it via an API endpoint managed within SAP BTP.

Methodology

Phase 1: Learning (From Video)

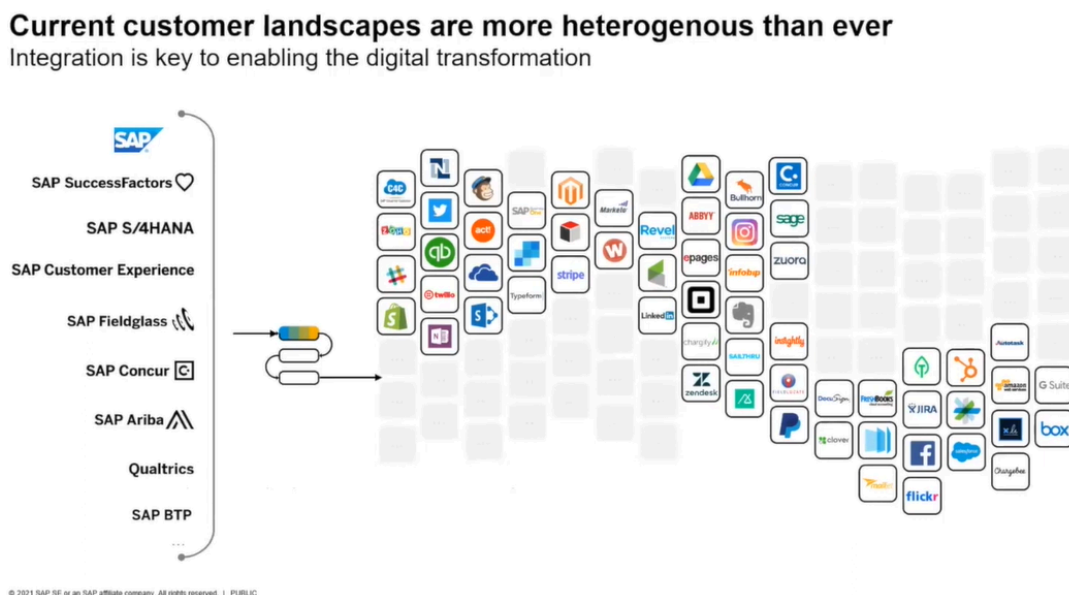


Figure 1.1 Customer Landscape

Typical customer landscapes are more heterogeneous than ever, and they are also changing very quickly. A combination of legacy on-premise solutions and cloud-based applications from a multitude of vendors are leveraged. This underlines how important integration is.

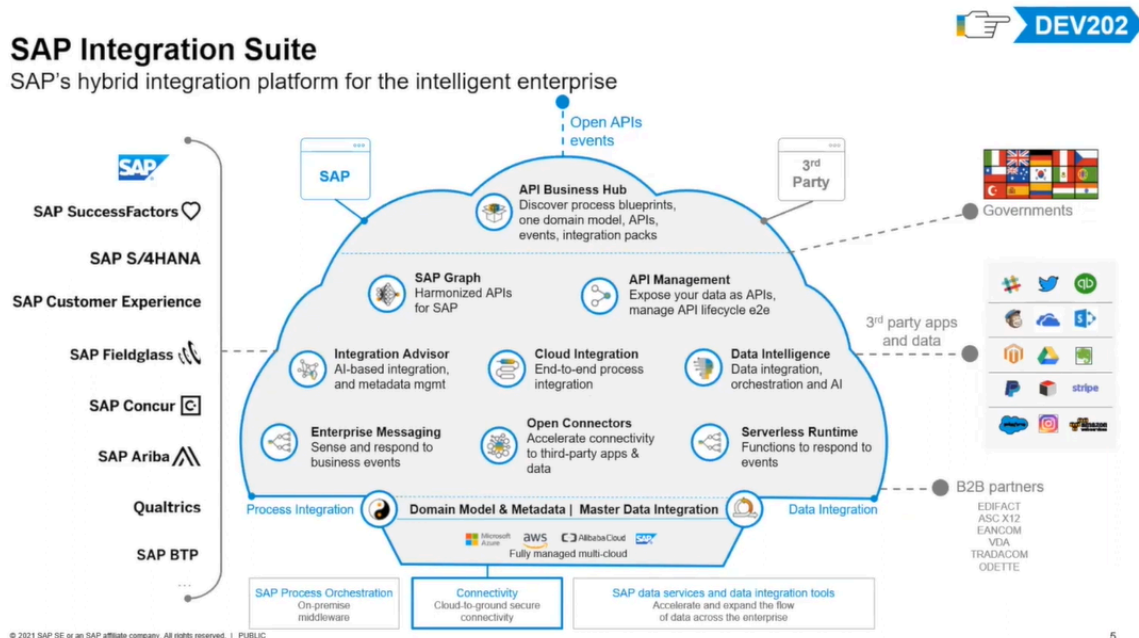


Figure 1.2 SAP Integration Suite

The integration suite is a versatile, hybrid and enterprise grade integration platform. It simplifies and accelerates integration projects and helps organizations to quickly transition to intelligent enterprises. It is composed of four main capabilities; cloud integration for end-to-end process integration, API management to design, deliver, and manage APIs, open connectors to simplify non-SAP connectivity, and the integration advisor to accelerate the interface design with the help of measuring learning capabilities. The integration suite with all capabilities is running on different hyperscaler environments such as AWS, Azure, Alibi Cloud, and Google Cloud. Gardner recognized the integration suite as one of the most comprehensive integration platforms, ranking it as a leader in the 2020 magic weapon for enterprise bypass.

Profit from quick time to value through out-of-the-box integration content

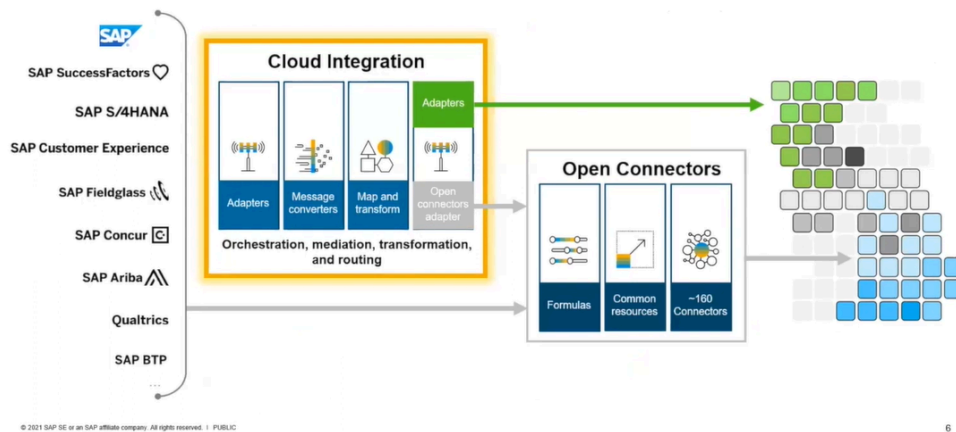


Figure 1.3 Bridge the two worlds of SAP and non-SAP Applications

Cloud integration is used as a middleware solution for the last-mile to non-SAP cloud applications, the open connectors capability can jump in. Cloud integration is the preferred option especially in scenarios which profit from quicktime to value through out of the box integration content from the SAP API business hub. For any integration where mediation, transformation, and routing are helpful.

Profit from quick time to value through out-of-the-box integration content

SAP API Business Hub

Search, discover, experience, and consume to right integration content from SAP and partners for your digital transformation projects

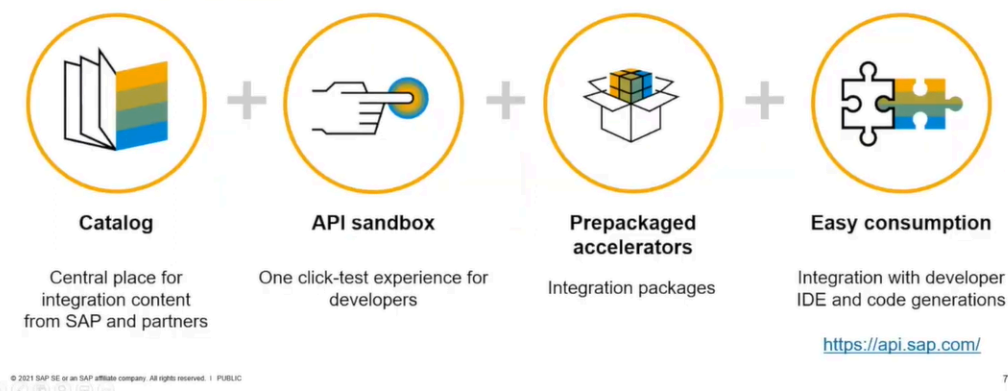


Figure 1.4 Steps to consume right SAP integration content

Customers can leverage pre-built integration packs from SAP and also from partners to get out of the box integrations for SAP to SAP and SAP to non-SAP integration scenarios, up and running quickly.

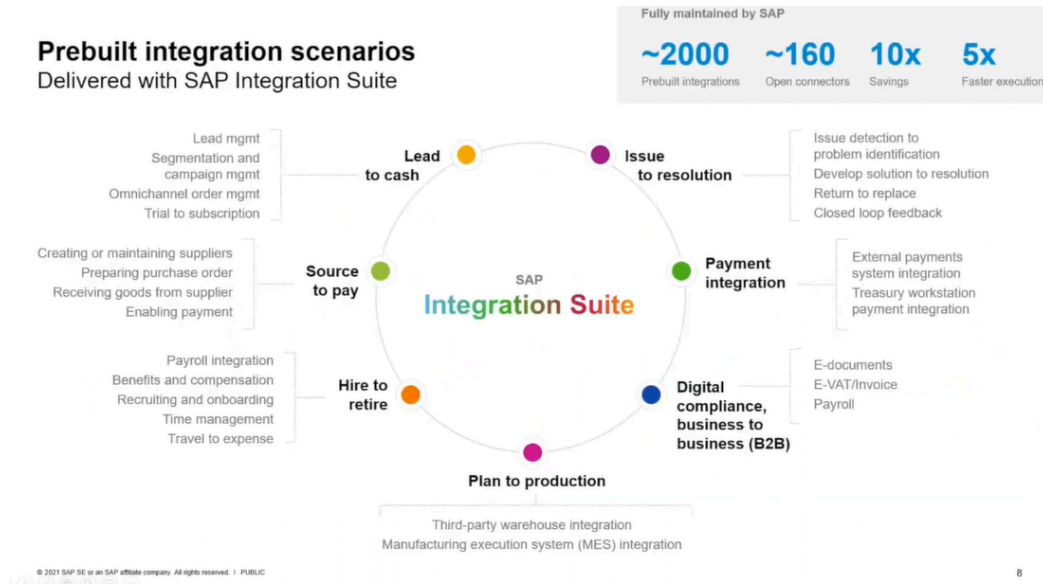


Figure 1.5 Prebuilt Integration Scenarios

There are approximately 2000 pre-built integration flows and value mappings, and around 160 open connectors are available at SAP API business hub. Integration content across all different critical core business processes are delivered as for leads to cash, source to pay at many others. More than 50% percent of the content is first up to non-SAP integration integration scenarios. Messages through pre-built SAP to SAP integration packages from the API business hub are free of charge and SAP provides full support and updates of the available content which reduces the effort required when application interfaces change. It also supports protocol-based adapters (SOAP, OData, JDBC) and application-specific adapters (Salesforce, SuccessFactors).

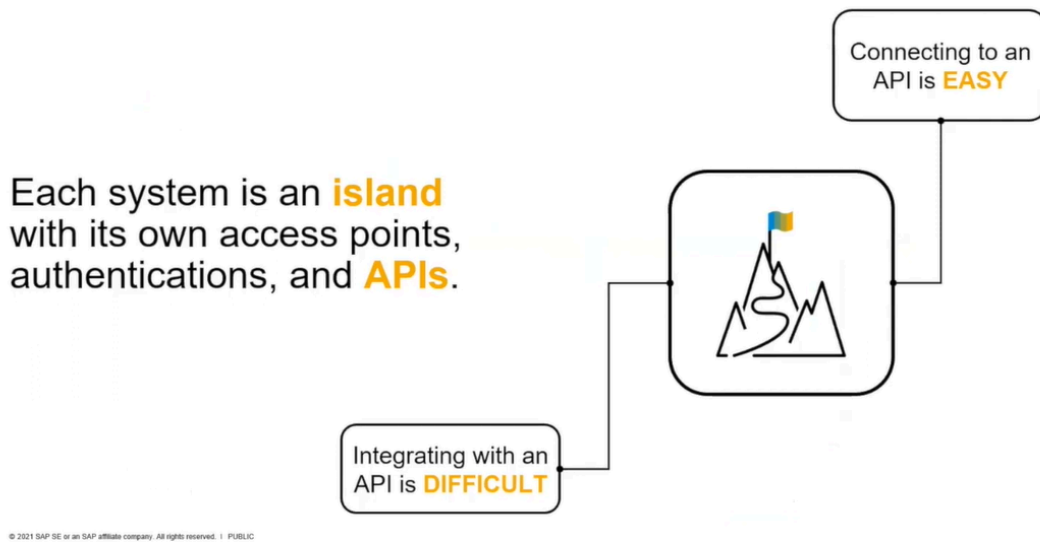
Digital compliance – E-government integration cloud



Figure 1.6 Example of SAP Integration around the world

Challenges of API-based integration

Each system is an **island** with its own access points, authentications, and **APIs**.



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Figure 1.7 Technical complexity of connecting to different third-party apps

Every non-SAP system acts like an “island” with unique authentication methods, error formats, and search syntaxes. Therefore, Open Connectors provide “normalized” features –meaning the platform harmonizes different behaviours into a standard format; the suite manages authentication complexities internally, meaning users do not need to master the specific OAuth flow for every individual application. It also creates consistency by mapping disparate error codes to standard HTTP status codes and translating a common query language into the specific endpoint’s syntax for search and pagination. Furthermore, the platform supports “Common Resources,” which allows users to define a canonical data model such as generic “Company” object that maps to fields across multiple systems like Salesforce or Microsoft Dynamics, ensuring consistent data usage across all integration projects.

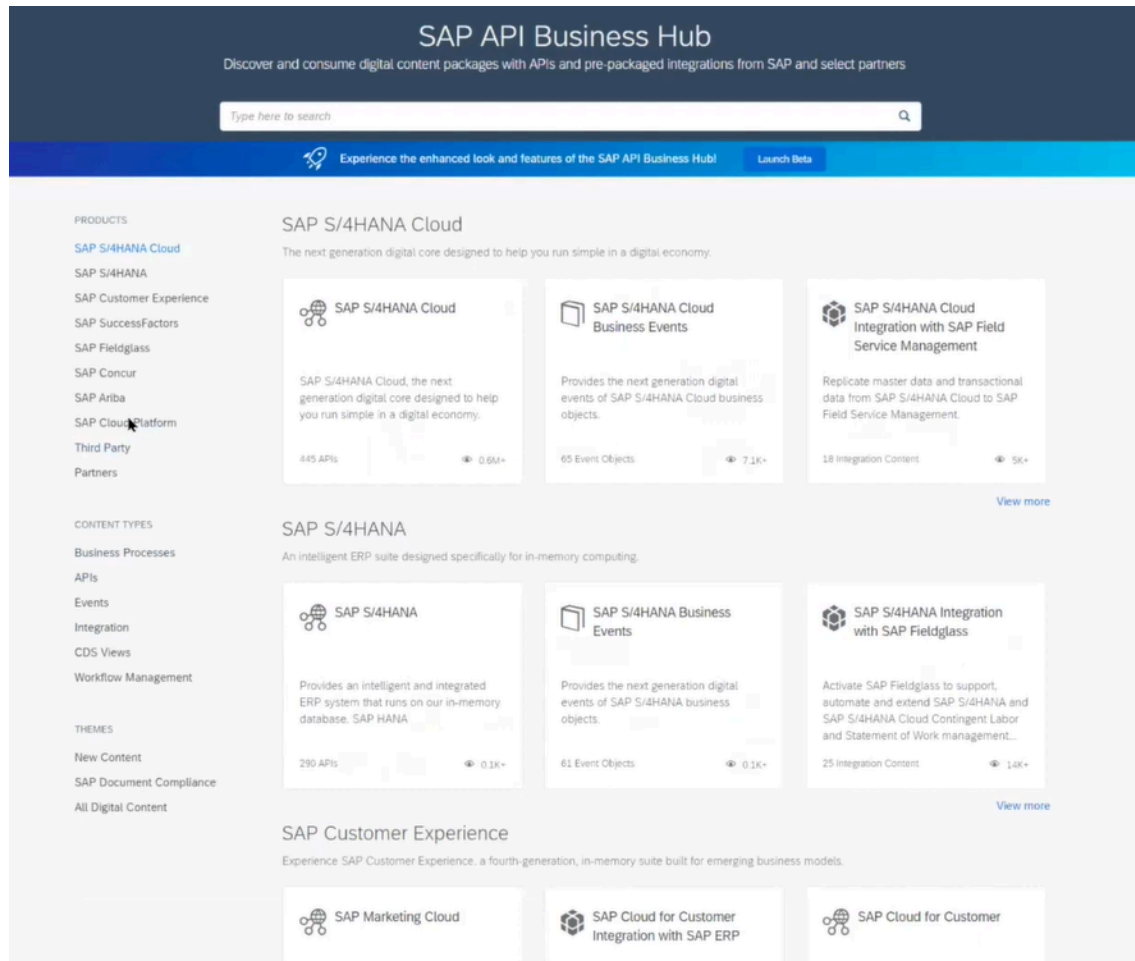


Figure 1.8 Salesforce Integration practical demonstrations

In the Salesforce integration demonstration, the efficiency of utilizing pre-packaged content available on the SAP API business hub is highlighted. Rather than building a connection from scratch, a “Salesforce to S/4HANA” scenario searches and identifies a ready-made integration package. The demonstration shows how the content can be easily copied directly into the cloud integration workspace. The process emphasizes a low-code approach, where the user simply needs to configure specific parameters such as the endpoint URL to get the integration up and running, eliminating the need for manual coding and significantly reducing implementation time.

Phase 2: Hands-On Application (BambooHR Integration)

1. The Use Case: Connectivity to Non-SAP Applications using SAP Integration Suite
 - What is the SAP Integration Rule?

SAP Integration Suite is an enterprise-grade integration platform as a service (EiPaaS) that allows you to smoothly integrate on-premise and cloud-based applications and processes with tools and prebuilt content managed by SAP.

SAP Integration Suite combines the integration capabilities such as Cloud Integration (Process Integration), API Management, Integration Advisor, Trading Partner Management, Integration Assessment, Open Connectors, and Migration Assessment into a cohesive and simplified toolkit for enterprise integrations. To provide a comprehensive integration experience, these services are not available separately, but only as part of SAP Integration Suite.

- The Use Case

Using this integration scenario, design and execute an integration flow that reads information from a third-party software. Use *Open Connectors* to identify and establish connection to a third-party software, and fetch data from the third-party software via APIs. *Cloud Integration* is used to design a custom integration flow to fetch data from Open Connectors, and deploy the integration flow to a cloud-based runtime location. Finally, use an open API Client like Postman to securely invoke the deployed integration artifact to see that data is able to be fetched from a third-party software using SAP integration suite.

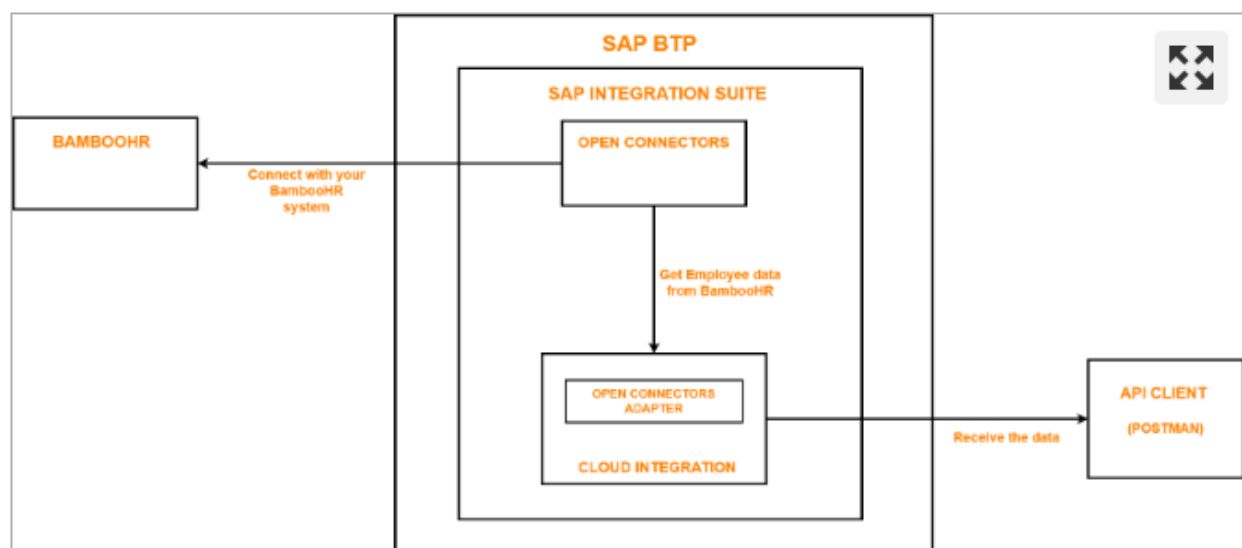


Figure 2.1.1 Example of Integration Flow

2. Preparing SAP Integration Suite Tenant

- Activate the capabilities

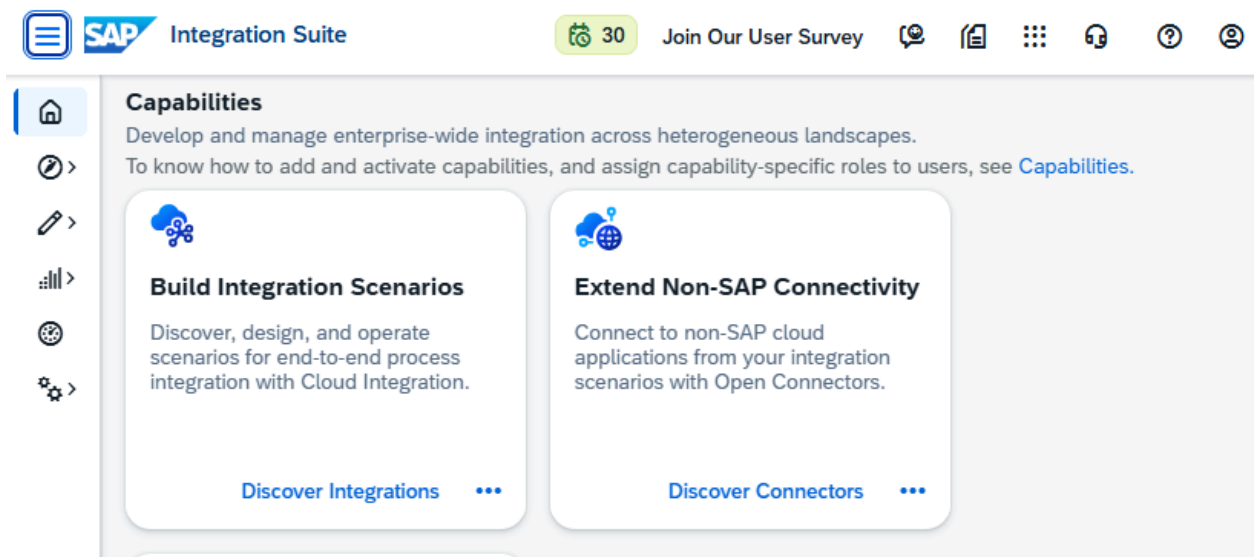


Figure 2.2.1 Activated capabilities

- Assign roles for the activated capabilities

Capability	Role Collection
Cloud Integration	PI_Administrator
Cloud Integration	PI_Business_Expert
Cloud Integration	PI_Integration_Developer
Open Connectors	OpenConnectors_User

Figure 2.2.2 Assign role collections to capabilities

- Create a service instance & a service key

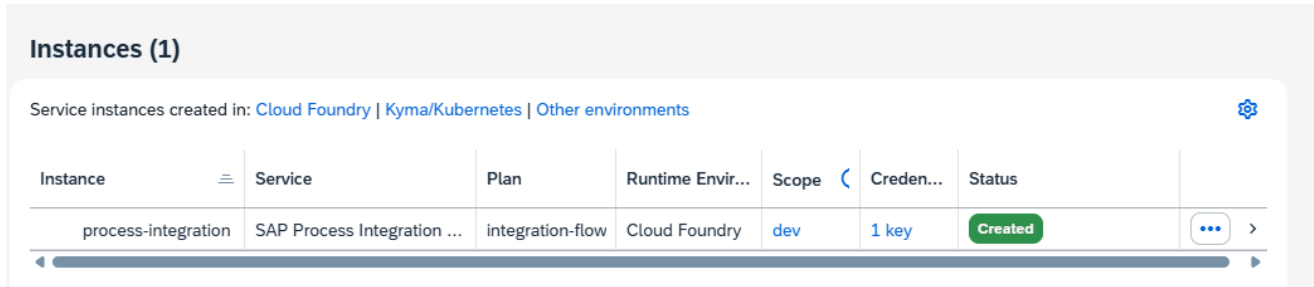


Figure 2.2.2 A service instance is created

An instance of SAP Process Integration Runtime service is created to access the endpoints after deploying the integration artifacts. That 1 service key under credential section is needed for the final step; the values **clientid** and **clientsecret**.

- Verify the activated capabilities

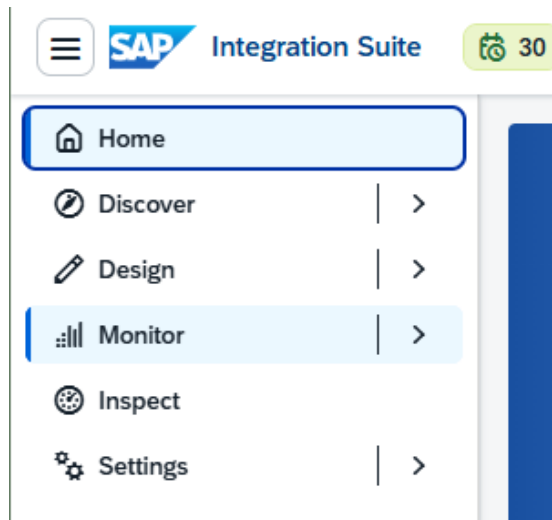
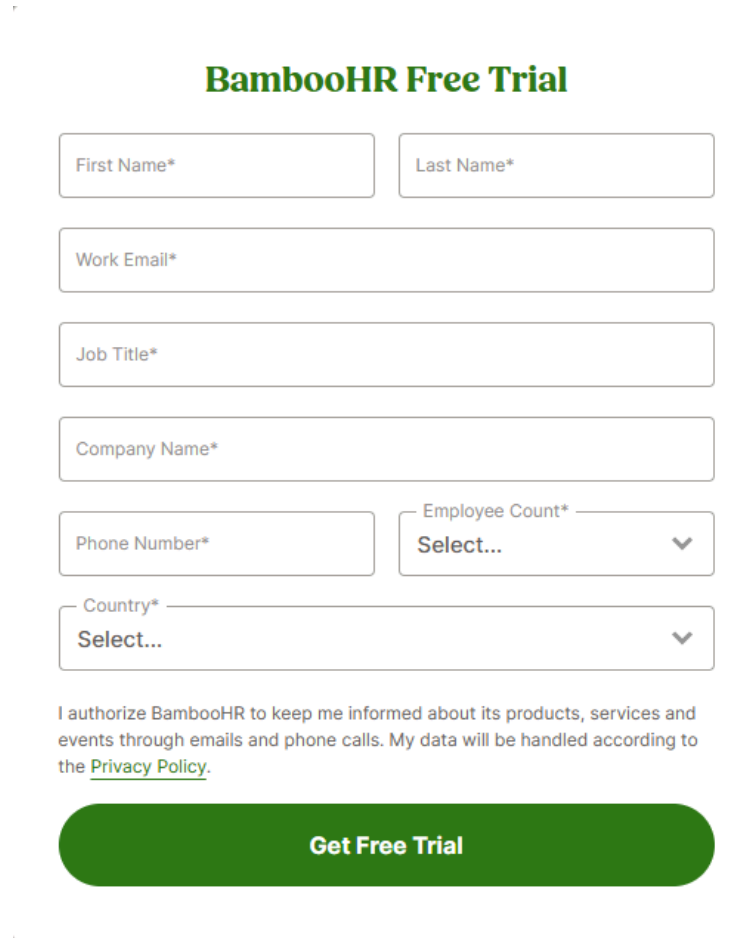


Figure 2.2.3 Capability-specific menu items are being added

3. Preparing Non-SAP Application's Trial Account

- Sign up for a free BambooHR Trial



BambooHR Free Trial

First Name* Last Name*

Work Email*

Job Title*

Company Name*

Phone Number* Employee Count*
Select... ▼

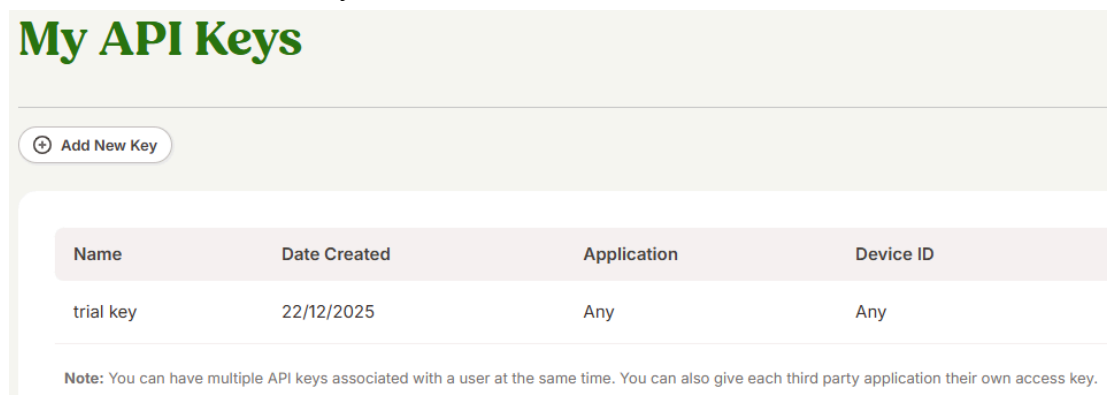
Country*
Select... ▼

I authorize BambooHR to keep me informed about its products, services and events through emails and phone calls. My data will be handled according to the [Privacy Policy](#).

Get Free Trial

Figure 2.3.1 Signup BambooHR free trial

- Generate an API Key



My API Keys

+ Add New Key

Name	Date Created	Application	Device ID
trial key	22/12/2025	Any	Any

Note: You can have multiple API keys associated with a user at the same time. You can also give each third party application their own access key.

Figure 2.3.2 API Key is generated

The API Key's address is a vital part of the next step in the mission, so it has to be copied.

4. Establish Connection Between BambooHR Trial & Open Connectors

- Access Open Connectors

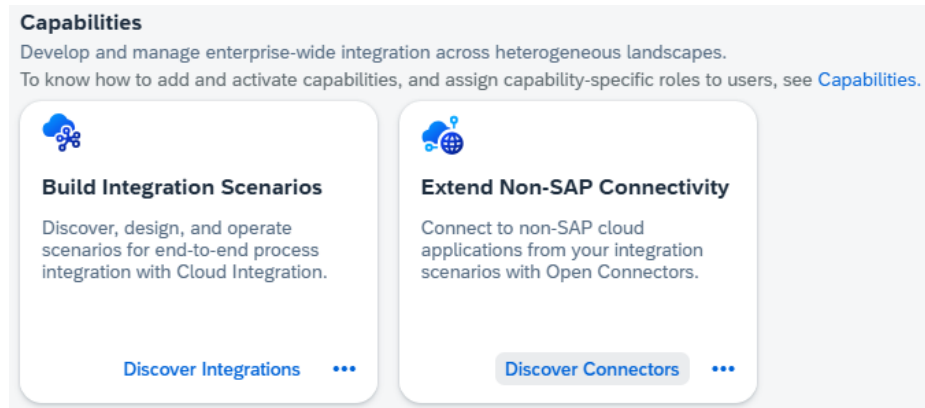


Figure 2.4.1 Discover Connectors through capabilities

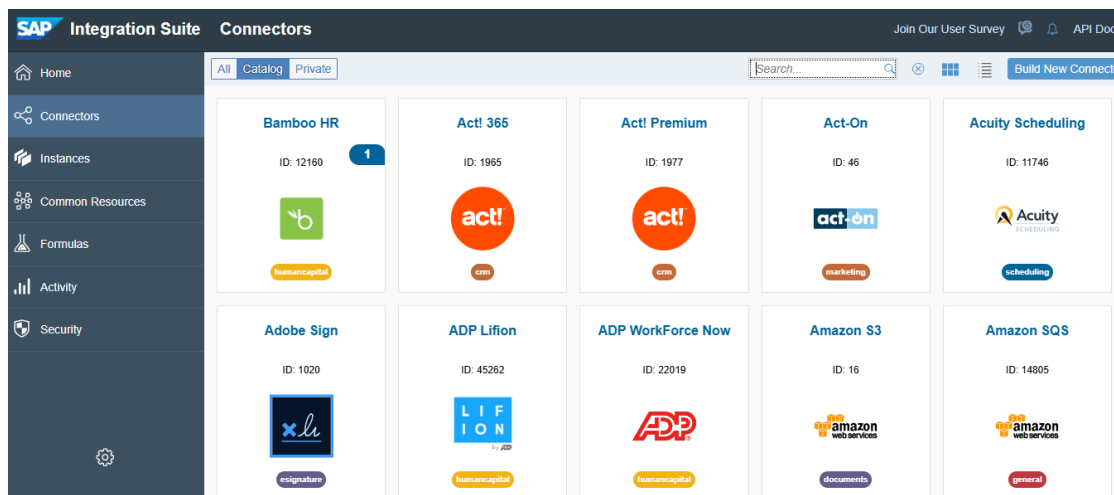


Figure 2.4.2 A catalog of various third-party applications that are supported by Open Connectors

- Authenticate with BambooHR's Domain Name & API Key



Figure 2.4.3 Choose API Docs of BambooHR

☐	Name	Id	Connector	Modified Date (UTC)	Enabled	
☒	bambooHR_connection	5399347	Bamboo HR	Mon Dec 22 2025 04:01:46 pm	Yes	API Docs

Figure 2.4.4 Created instance

- Test the API Docs

Curl

```
curl -X GET "https://api.openconnectors.trial.ap21.ext.hana.ondemand.com/elements/api-v2/employees" -H "accept: application/json" -H "Authorization: User myFUo1bZkQgrdzDRdI+Wjz18mb05nV51Svv6kvPzntk=, Organization e2ac47f0ccdb25e670f4aee27b4ed180, Element z3V0nSKY8nph1TQoi0zP4+mvMCjjzmPr4QrarnsQ2+U="
```

Request URL

https://api.openconnectors.trial.ap21.ext.hana.ondemand.com/elements/api-v2/employees

Server response

Code Details

200 Response body

```
[
  {
    "lastName": "Abbott",
    "twitterFeed": "@twitter",
    "workEmail": "cabbott@efficientoffice.com",
    "displayName": "Charlotte Abbott",
    "jobTitle": "Sr. HR Administrator",
    "facebook": "www.facebook.com",
    "pinterest": "www.pinterest.com",
    "linkedin": "www.linkedin.com",
    "instagram": "@instagram",
    "canUploadPhoto": 1,
    "division": "North America",
    "firstName": "Charlotte",
  }
]
```

Show raw Copy Download

Figure 2.4.5 Tested API Docs connection

- Create a Security Material in Cloud Integration

Security Material (1)						Filter by Name or Deployed By	Q	Create	Upload	Refresh	Settings
Name	Type	Status	Deployed By	Deployed On	Actions						
MyOCUser	User Credentials (Open Connectors)	Deployed	adriana04@graduate.utm.my	Dec 22, 2025, 22:40:17	Edit Delete						

Figure 2.4.6 Created a security material in Cloud Integration

- Copy the Request URL

Request URL

https://api.openconnectors.trial.ap21.ext.hana.ondemand.com/elements/api-v2/employees

Figure 2.4.7 The copied Request URL for Cloud Integration to invoke

5. Design an Integration Flow to Connect with Open Connectors

- Create an Integration Package and Integration Flow

Packages (1)							Search	↕
Name	Mode	Ve...	Created By	Created Date	Description	Ac...		
 CommunityChallenge	Editable	1.0.0	adriana04@graduate.utm.my	Mon, 22 Dec 2025 14:42:28 GMT	CommunityChallenge			>

Figure 2.5.1 An integration package named Community Challenge is created

☐	Name	Type	Version	Actions
☐	Non-SAP_Connectivity	Integration Flow	Draft	
	Created Locked by: adriana04@graduate.utm.my (you)			

Figure 2.5.2 Created an integration flow called Non-SAP_Connectivity

- Edit the integration flow

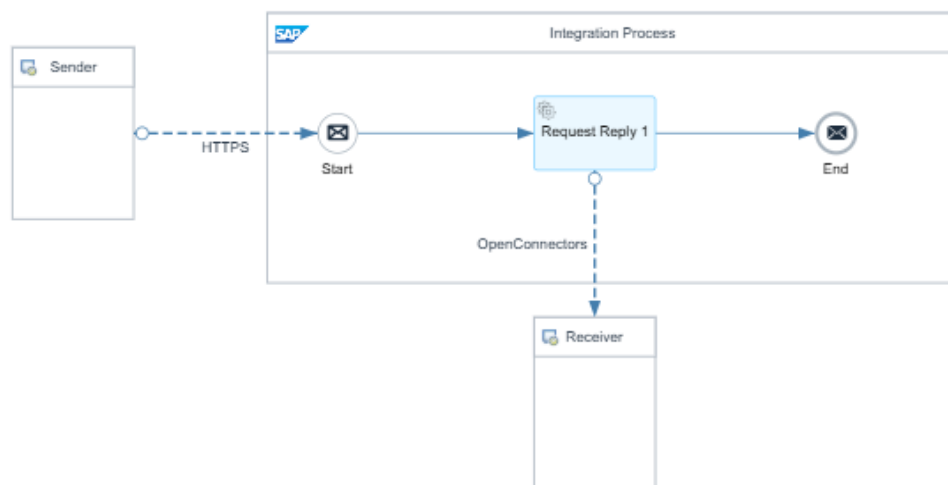


Figure 2.5.3 The designed integration flow is saved and ready to be deploy

Components used: Sender Adapter (HTTPS), Request-Reply step, Receiver Adapter (OpenConnectors). I used the Cloud Integration capability to design a flow that accepts an HTTPS request and routes it to the BambooHR system.

- Deploy the integration flow

Integration Flow					
General	Runtime Configuration	Error Configuration	References	Externalized Parameters	Deployment Status
Deployment Status: Deployed Navigate to Manage Integration Content Runtime Status:  Started Deployed By: adriana04@graduate.utm.my Deployed On: Dec 23, 2025, 01:21:07 Deployed Version: 1.0.0					

Figure 2.5.3 The integration flow is successfully deployed

6. Test The Integration Scenario

- Get The Endpoint of The Deployed Integration Flow

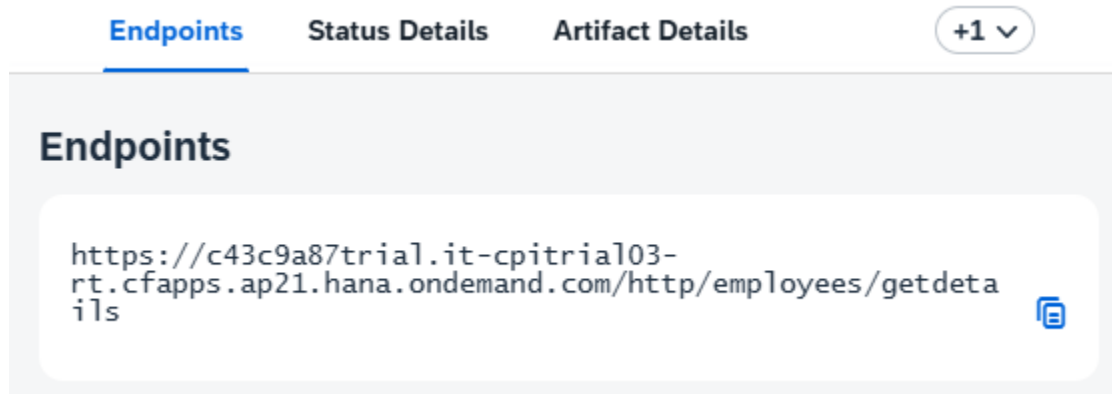


Figure 2.6.1 Copied the endpoint address of the deployed integration flow

- Test the Integration Scenario

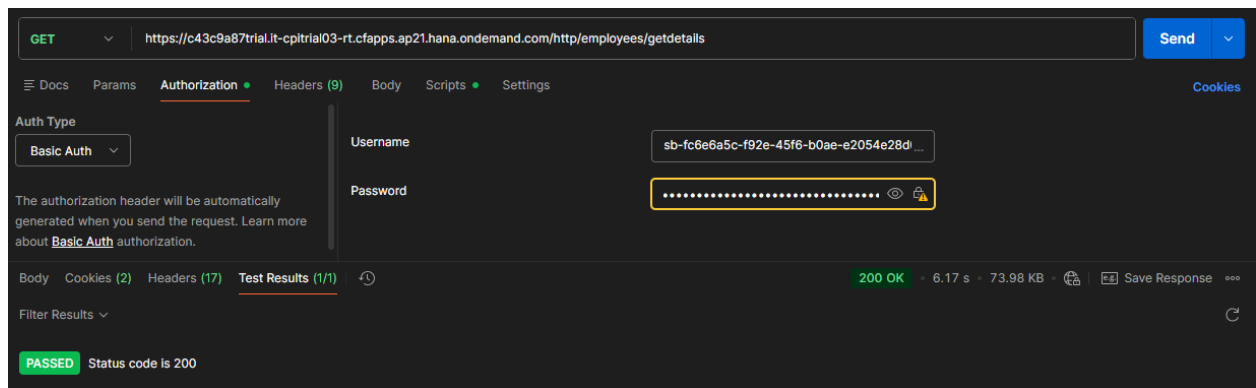


Figure 2.6.2 Tested integration design using Postman API Testing Platform

I utilized Postman to simulate external application requests, using Client ID and Client Secret for Basic Authentication.

Challenges & Troubleshooting

- Instance Configuration (Error 404): Initially, the BambooHR instance failed to authenticate because the “Company Name” field included the full URL rather than the subdomain (universititeknologimalaysia). I learned that Open Connectors builds the URL dynamically, requiring precise input.

- Path Duplication (Error 500): I encountered an Internal Server Error where the API path was duplicated (e.g., /employees/employees). Resolved this by correcting the Resource Path in the Receiver Adapter and ensuring the “Draft” version of the flow was not being deployed over the active version.

Key Learnings

I learned that SAP Integration Suite acts as a powerful “translator” between systems. The experience highlighted the importance of exact URI configuration and how to use the Message Processing Log (MPL) to diagnose backend failures (500 vs. 404 errors).

Implied SAP BTP into Subsystem

Subsystem: Course Details Module

In the tutorial, I learned how to decouple the frontend from the data source by using an API middleware to fetch employee records from BambooHR. Similarly, in our Course Registration System, the *Course Details Module* that I’m working on currently should not ideally store static syllabus data, instead it should act as a consumer that fetches real-time course attributes (credit hours, prerequisites, quotas) via APIs.

By applying the concepts of RESTful API integration and secure authentication I practiced on BTP, I can ensure my Course Details module is designed to retrieve data dynamically, similar to how I retrieved the JSON employee data in this demo.

References

SAP Community. (2021, February 8). *Integrate Non-SAP Applications with SAP Integration*

Suite | *SAP TechEd for SAP Community* [Video]. YouTube.

<https://www.youtube.com/watch?v=H18cSAh5-hc>